

FELICIA NONTOKOZ PHIRI
BRIGHTLEARN DATA ANALYTICS
RESEARCH ASSIGNMENT 1

SECTION A : DATA FUNDAMENTALS

① A Database is a digital repository for storing, managing and securing organized collections of data.

It allows users to easily store, retrieve, update and analyze information.

Different types of databases store data in different ways. For example, relational databases store it in defined tables with rows and columns, while nonrelational databases can store it as a variety of data structures.

Why databases are essential:

- Databases play a critical role in almost every industry today. They store large amounts of data in a structured way, making it easy to find and use.
- They promote data accuracy and consistency by reducing duplication and errors by keeping data centralized.
- Help organizations analyze data and make informed decisions.
- Protection of sensitive information through access controls and encryption.

TWO REAL WORLD EXAMPLES

① Banking Systems

Banks store customer details, account balances and transaction records in databases to process payments and withdrawals securely.

② Social Media Platforms

Applications like Facebook store user profiles, posts, messages, and interactions in databases.

② WHAT IS A DATA WAREHOUSE? HOW DOES IT DIFFER FROM A REGULAR DATABASE? WHY WOULD A COMPANY USE A DATA WAREHOUSE?

A data warehouse is a specialized type of database designed to store large amounts of historical data from different sources in one central place, so it can be analyzed and used for reporting and decision-making.

Unlike regular databases (which handle daily operations like transactions), a data warehouse is used mainly for analysis, trends and business insights.

Data warehouse typically uses extract, transform, and load (ETL) or extract, load, and transform (ELT) processes to clean, prepare and organize data for business intelligence (BI).

A company uses a data warehouse to turn large amounts of data into meaningful insights that support better decision making.

Companies often have data in different systems (sales, finance, marketing). A data warehouse brings all this data together into one central place, making it easier to analyze.

Data warehouse keeps data for many years, helping companies to track long-term trends.

③ WHAT IS A DATA LAKE? EXPLAIN THE CONCEPT AND DESCRIBE HOW IT DIFFERS FROM A DATA WAREHOUSE. WHEN WOULD A BUSINESS PREFER A DATA LAKE OVER A DATA WAREHOUSE?

Data lake is a centralized repository that ingests and stores large volumes of data in its original form. The data can be processed and used as a basis for a variety of analytic needs.

Data lakes and data warehouses are similar in that they both store and process data, each have their own specialties and therefore their own use cases.

A data lake captures both relational and non-relational data from a variety of sources.

- Business applications, mobile apps, IoT devices, social media, or streaming - without having to define the structure or schema of the data until it is read. Schema-on-read ensures that any type of data can be stored in its raw form.

By contrast, a data warehouse is relational in nature. The structure or schema is modeled or predefined by business and product requirements that are curated, conformed, and optimized for SQL query operations. While a data lake holds data of all structure types, including raw and unprocessed data, a data warehouse stores data that has been treated and transformed with a specific purpose in mind, which can be then used to source analytic or operational reporting.

A business would prefer a data lake over a data warehouse when it needs flexibility, raw data storage, and advanced analytics rather than structured reporting.

DATA LAKE

- Raw, unstructured
- Flexible
- Big data & AI
- Store now, use later

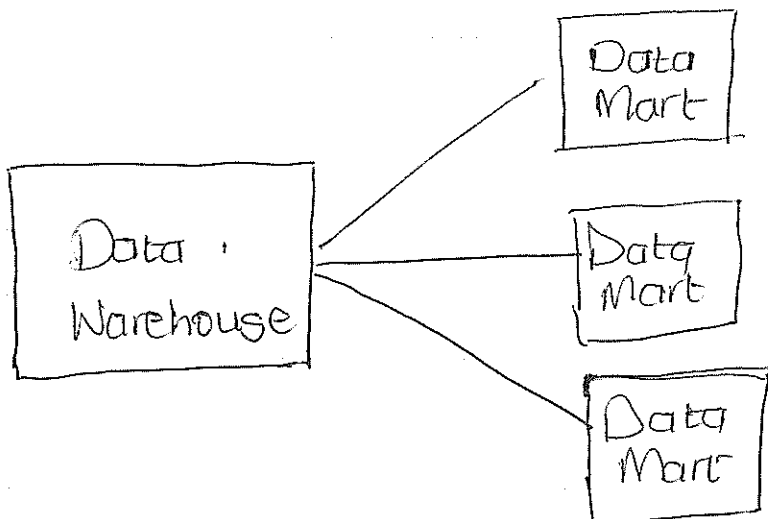
DATA WAREHOUSE

- Clean, structured
- Organized
- Reporting & BI
- Use immediately

④ WHAT IS DATA MART? HOW DOES IT RELATE TO A DATA WAREHOUSE AND WHO TYPICALLY USES IT?

A data mart is a subset of a data warehouse focused on a particular line of business, department or subject area. Data marts can improve team efficiency, reduce costs and facilitate smarter tactical business decision-making in enterprises.

Data mart is built from an existing data warehouse (or other data sources) through a complex procedure that involves multiple technologies and tools to design and construct a physical database.



⑤ EXPLAIN DIFFERENCE BETWEEN STRUCTURED, SEMI-STRUCTURED, AND UNSTRUCTURED DATA.

- Structured data - is organized in a clear, predefined format. It can include both quantitative data and qualitative data.

Example : Financial reports

- Unstructured data - does not have a predefined format. Unstructured datasets are typically large.
Example : Emails, Social media posts.

- Semi-STRUCTURED - does not have a predefined data model. However it uses metadata to identify specific data characteristics.

Example : Java Script Object Notation.

⑥ Data Modeling is the process of creating a visual representation of either a whole information system or parts of it to communicate connections between data points and structures

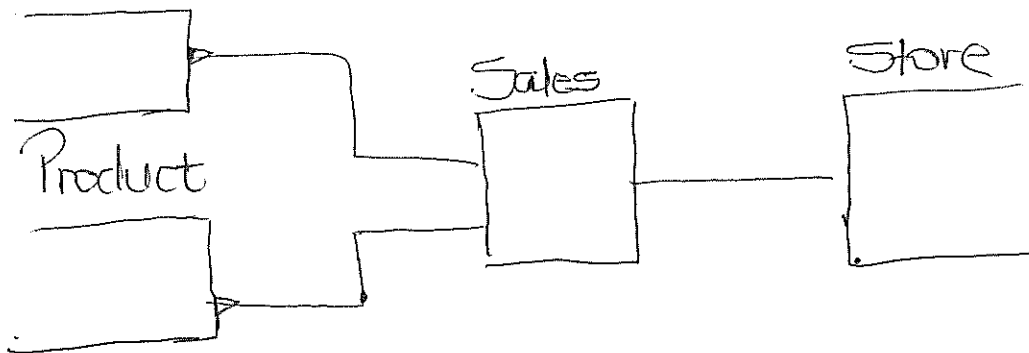
Data models are built around business needs. Rules and requirements are defined upfront through feedback from business stakeholders so they can be incorporated into the design of a new system or adapted in the iteration of existing one. Data modeling helps designers understand how data should be organized before building the system, reducing confusion later.

Data modeling makes systems more efficient and improves data quality.

TYPES OF DATA MODELS.

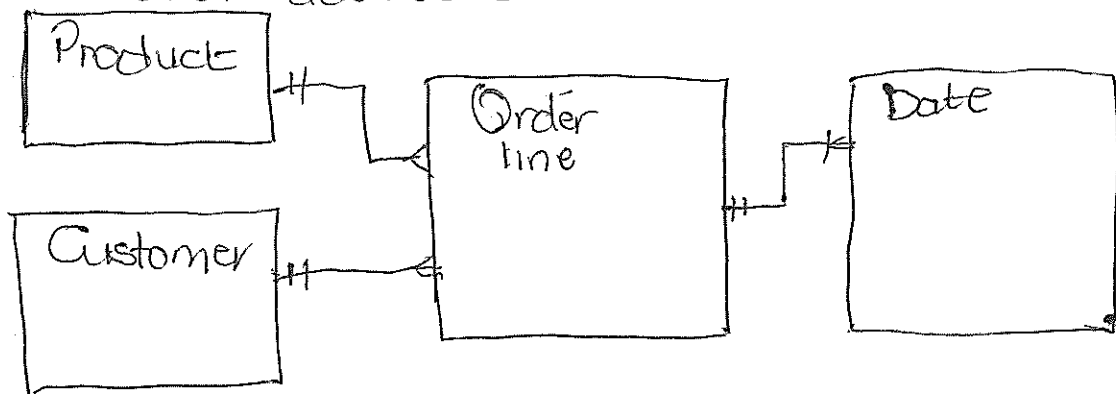
- Conceptual data Models

They are also referred to as domain models and offer a big picture view of what the system will contain, how it will be organized, and which business rules are involved.



- Physical data models

They provide a schema for how the data will be physically stored within a database. As such, they are the least abstract of all. They offer a finalized design that can be implemented as a relational database.



7. Data Mining is the process of discovering meaningful patterns, relationships and insights from large volumes of data. It draws on techniques from statistics, machine learning and data management to surface signals that are not immediately obvious through simple queries or reporting.

Data mining is typically executed on big data analytics platforms that can handle large volumes of data efficiently and reliably.

REAL WORLD DATA MINING APPLICATIONS

- Healthcare

Data mining play an increasingly important role in improving patient outcomes. Predictive models are used to identify patients at higher risk of complications, enabling earlier intervention and more proactive care.

- Financial Institutions

Rely heavily on data mining to manage risk and protect against fraud. Anomaly detection models analyze transaction data in real time to identify unusual behaviour that may indicate fraudulent activity. Many organizations accelerate this capability using purpose-built solutions for fraud detection.

8. Database Management System

-Is a combination of tools, processes, and software used to collect, store, organize, protect, and manage data throughout its lifecycle.

FOUR POPULAR DBMS

- MySQL

A widely used open-source relational DBMS. Known for being fast, reliable and easy to use. Commonly used in web applications and websites.

- Oracle Database.

Is a popular and fully featured Relational Database management system developed by Oracle Corporation. It is known for its reliability, scalability, and extensive features. This makes it a top choice for businesses needing to manage and retrieve large amounts of data efficiently.

- Microsoft SQL Server.

A DBMS developed by Microsoft for business and enterprise use. Integrates well with other Microsoft products. Strong in data analysis and reporting.

- MongoDB.

A popular NoSQL DBMS that stores data in flexible, JSON-like documents.

Good for handling unstructured or semi-structured data.

Widely used in Modern apps and big data systems.

9. WHAT IS ETL?

A core data integration process used to gather data from multiple sources, clean and standardize it, and load it into a data warehouse, data lake or other target system for analytics and reporting.

- (i) Extract data from legacy systems.
- (ii) Cleanse the data to improve data quality and establish consistency.
- (iii) Load data into a target database.

10. COMMON DATA STORAGE FORMATS

(i) CSV (Comma-Separated Values).

A simple text-based format used for storing tabular data, where each line represents a row and values are separated by commas.

- Widely used for data exchange between applications, spreadsheets and databases.

(ii) JSON (JavaScript Object Notation).

A lightweight data interchange format that is easy for humans to read and write, and easy for machines to parse and generate.

- Used for API's and web services

(iii) XML (Extensible Markup Language).

- A markup language that defines rules for ~~ee~~ encoding documents in a format that is both human-readable and machine-readable.
- Often used for data interchange between systems, especially in web services and configuration files. Highly structured and supports complex data hierarchies.

(iv) Avro.

A row oriented data serialization framework that provides a compact binary format.

- Commonly used in data-intensive applications, particularly in the Hadoop ecosystem.

Supports schema evolution, allowing for changes in data structure over time without breaking compatibility.

SECTION B : ARTIFICIAL INTELLIGENCE & MACHINE LEARNING.

11. Artificial Intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.

Strong AI

Is essentially AI that is capable of human-level, general intelligence. In other words, its just another way to say "artificial general intelligence".

Weak AI

Refer to the narrow use of widely available AI technology, like machine learning or deep learning, to perform very specific tasks, such as playing chess, recommending songs, or steering cars. Also known as Artificial Narrow Intelligence (ANI).

Examples: Apple's Siri, social media chatbot.

12. Machine Learning, is the subset of artificial intelligence (AI), focused on algorithms that can "learn" the patterns of training data and, subsequently, make accurate inferences about new data. This pattern recognition ability enables machine learning models to make decisions or predictions without explicit, hard-coded instructions.

HOW DOES MACHINE LEARNING DIFFER FROM TRADITIONAL PROGRAMMING.

- The key difference is how the system learns and produces results.

Traditional Programming $\Rightarrow$ You tell the computer how to solve the problem.

Machine Learning $\Rightarrow$ The computer learns how to solve the problem from data.

THREE MAIN TYPES OF MACHINE LEARNING.

Supervised learning trains a model to predict the "correct" output for a given input. It applies to tasks that require some degree of accuracy relative to some external "ground truth", such as classification or regression.

- Unsupervised Learning trains a model to discern intrinsic patterns, dependencies and correlations in data. Unlike in supervised learning, unsupervised learning tasks don't involve any external ground truth against which its outputs should be compared.

- Reinforcement learning (RL) trains a model to evaluate its environment and take an action that will garner the greatest reward. RL scenarios don't entail the existence of a singular ground truth, but they do entail the existence of "good" and "bad".

13. Deep Learning is a type of machine learning that uses artificial neural networks to learn from data, similar to the way we learn. Artificial neural networks are inspired by the human brain, and they can be used to solve a wide variety of problems, including image recognition, natural language processing, and speech recognition.

Neural networks are made up of layers of interconnected nodes, and each node is responsible for learning a specific feature of the data.

14. Natural Language Processing is a subfield of Artificial Intelligence (AI) that focuses on enabling computers to understand, interpret and generate human language in a meaningful way. It combines computational linguistics with machine learning and deep learning techniques to process and analyze text and speech data effectively.

NLP is already part of everyday life for many, powering search engines, prompting chat bots for customer service with spoken commands, voice operated GPS systems and Apple's Siri.

15. Large Language Models (LLMs) are a category of deep learning models trained on immense amounts of data, making them capable of understanding and generating natural language and other types of content to perform a range of tasks.

LLM's are built on a type of neural network-architecture called a transformer which excels at handling sequences of words and capturing patterns in text.

LLM's represent a major leap in how humans interact with technology because they are the first AI system that can handle unstructured human language at scale, allowing for natural communication with machines.

TWO WELL KNOWN LLM'S

(i) Google's Gemini assistant.

- Used for search assistance, content generation and reasoning tasks.

(ii) Chat GPT.

- Used for answering questions, writing content, tutoring and coding.
- Helps with conversations, essays and problem-solving.

16. Generative AI is artificial Intelligence (AI) that can create original content such as text, images, video, audio or software code in response to a user's prompt or request.

HOW GENERATIVE AI WORKS.

- (i) Training, to create a foundation model that can serve as the basis of multiple gen AI applications.
- (ii) Tuning, to tailor the foundation model to a specific gen AI application.
- (iii) Generation, evaluation and retuning, to assess the gen AI application's output and continually improve its quality and accuracy.

TOOLS THAT USE GEN AI.

- (i) GitHub - Generates code suggestions
- (ii) DALL-E - Creates images from text descriptions.
- (iii) ChatGPT - Generates human-like text based on prompts.

17. Training Dataset - is information that is used to teach a machine learning model how to make predictions, recognize patterns or generate content. After an algorithm processes a vast amount of data, they are considered to be "trained" and usable for many applications.

Dataset could be made up of spreadsheets, video footage, web pages, PDFs or any type of data.

The quality of the training data directly affects how well the model performs.

If the training data is poor, several problems can occur, including incorrect predictions and misleading insights.

18. Computer vision. is the science and engineering of enabling computers to interpret and understand the visual world. Much like human eyes and brain work together to make sense of visual input, computer vision allows machines to analyze images and video, extract meaningful information.

TWO REAL-WORLD USE CASES OF COMPUTER VISION

In agriculture, drones equipped with cameras and AI analyze crop health, detect pests and estimate yields. Farmers can monitor vast fields more efficiently, optimize irrigation and reduce waste.

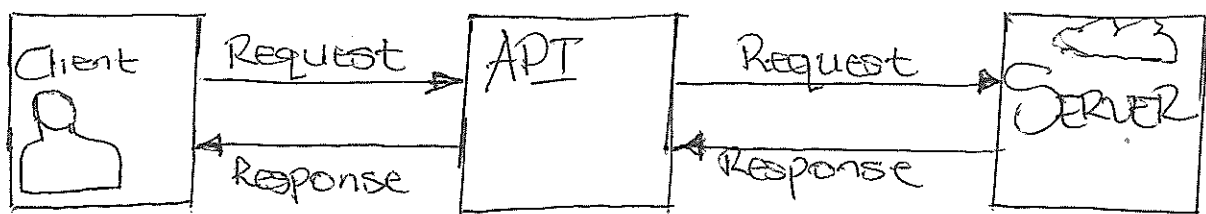
In retail, AI vision powers cashier-less stores where customers simply pick up items and walk out - the system charges them automatically.

SECTION C : APIs, TOOLS & DEVELOPER CONCEPTS.

19) What is API.

APIs are the invisible backbone of modern software development. They enable applications and systems to communicate and share data efficiently.

They enable seamless interaction between applications, platforms and services. Power everyday feature such as weather updates, payments and e-commerce checkouts.



APIs are crucial and critical to building scalable, flexible, and connected systems, because of their reusability, efficiency and integration.

20. What is API key.

An API key is a unique code (usually a long string of letters and numbers) used to identify and authenticate a user or application when accessing an API.

Purpose of an API key

- * Authentication : Confirms who is making the request.
- * Access control : Determines what data or services can be used.

*Tracking Usage : Monitors how much the API is being used.

*Security : Helps prevents unauthorized access.

How should API keys Be Stored?

- ① Keep it secret
- ② Use Environment variables
- ③ Use secure storage
- ④ Restrict permission.
- ⑤ Rotate keys regularly.

What happens if API key is exposed?

- Unauthorized user can access your services
- You may be charged for their usage.
- Data could be stolen or misused.

21. REST stands for Representational State Transfer and follows the constraints of REST architecture allowing interaction with RESTful web services.

It defines a set of functions (GET, PUT, POST, DELETE) that clients use to access server data.

- GET (retrieve a record)
- PUT (update a record)
- POST (create a record)
- DELETE (delete the record).

22. What is JSON (Java Script Object Notation)?

Is a lightweight data interchange format.

It is language-independent, which means it doesn't require a specific programming language.

JSON is commonly used in RESTful APIs to send and receive structured data between clients and servers.

Example of JSON.

```
{  
  "name": "Nonto",  
  "age": 30,  
  "email": "nontophiri@gmail.com"  
}
```

• `{ }` → defines a JSON object

• Keys (e.g. "name", "age") → describe data.

23. Prompt Engineering.

Is the process of designing and refining the prompts you give to an AI system so that it produces accurate, useful and relevant responses. The basic rule is that good prompts equal good results.

Prompt engineering is an important skill when working with AI tools as it helps in crafting better questions for better AI answers.

Examples.

- Weak Prompt: "Tell me about marketing"
- Strong Prompt: "Explain digital marketing strategies for small businesses in 5 bullet points with examples."
- Weak Prompt: "Help me write a Facebook Post"
- Strong: "Write a Facebook post about handbags with a catchy hook."

24. MAP Function is a built-in higher-order function that allows you to apply a given function to each item in an iterable (such as a list, tuple or set) and returns an iterator of the results.

• What Does it Do?

- Takes a collection of data. (e.g list or array)
- Applies a function (operation) to each item.
- Returns a new collection with updated value

Example in Python.

```
numbers = [1, 2, 3, 4, 5]
```

```
# Function to double each number
```

```
def double(x):
```

```
    return x * 2.
```

Output : { 2; 4; 6; 8; 10 }.

Why its useful in Data Processing

- Quickly transforms large datasets.
- Make code shorter and cleaner.
- Common in data analysis and pipelines.

25. Cloud Computing is the delivery of computing services over the internet instead of using local computers or servers.

3 Main Cloud Service Models

- IaaS (Infrastructure as a Service). e.g Amazon Web Service.
- PaaS (Platform as a Service). e.g Google App Engine.
- SaaS (Software as a Service) e.g Microsoft 365

26. Version Control System is a tool that helps you track changes to files over time. It tracks and records changes to the codebase, maintaining a structured project history.

Git is a distributed version of VCS used to manage and track changes in code and data projects.

-Uses

- Collaboration
- History Tracking.
- Branching
- Backup
- Speed & Flexibility.

(i) Commit : is a saved snapshot of your project at a specific point in time.

(ii) Branch is a separate version of your project where you can work independently.

27. Jupyter Notebook. is an open source web application that you can use to create and share documents that contain live code, equations, visualizations, and text. Most commonly used with python. It can also support other languages like R and Julia, but Python is the most popular.

Why is it Popular Among Data Analysts & Scientists

- (i) Interactive and Easy to Use.
- (ii) Combines Code and explanation.
- (iii) Great for Data Visualization.
- (iv) Supports Data Analysis Libraries.
- (v) Ideal for Experimentation.
- (vi) Easy to Share.

28. Python

Python is a programming language that is widely used in web applications, software development, data science, and machine learning (ML). Developers use Python because it is efficient and easy to learn and can run on many different platforms.

Why Python is Popular for Data Analytics and AI.

- Easy to Learn and use.
- Large Community
- Powerful Libraries
- Versatile
- Integration

Four Common Python Libraries

(i) Pandas

- Used for data manipulation and analysis.
- Works with tables like Excel.

(ii) NumPy

- Used for fast mathematical operations.
- Handles arrays and numerical data efficiently

(iii) Scikit-learn

- Used for machine learning tasks.
- Includes tools for classification, regression and clustering.

SECTION D : DATA ETHICS, GOVERNANCE AND CAREERS.

29. Data Privacy

Also called "information privacy", is the principle that a person should have control over their personal data, including the ability to decide how organizations collect, store and use their data.

The average organization today collects a massive amount of consumer data. Organizations have a responsibility to ensure the privacy of this data - not out of the goodness of their hearts, but as a matter of regulatory compliance, security posture and competitive advantage.

What is GDPR?

The General Data Protection Regulation is a data protection law introduced by the European Union (EU) in 2018.

- It is designed to:
 - Protect individuals' personal data.
 - Give people more control over their information.
 - Regulate how organizations collect and process data.

Key features of GDPR.

- Consent
- Right to Access
- Right to be Forgotten
- Data Protection
- Breach notification

Applies to all countries in the European Union.

(30) DATA ETHICS.

- Refers to the moral principles and guidelines that govern how data is collected, used, shared and managed responsibly.

EXAMPLES OF UNETHICAL USE OF DATA OR AI.

(i) Unauthorized Data Collection.

- Violates privacy rights and breaks trust between users and organizations.

(ii) Biased AI Systems.

- AI trained on biased data that leads to unfair decisions.
- Leads to discrimination and inequality. Produces unfair outcomes.

(iii) Misuse of Personal Data for Manipulation.

- Using data to influence people's behaviour unfairly.
- Manipulates people's opinions without transparency.
- Exploits personal information.

(31) Data Governance.

Is the data management discipline that focuses on the quality, security and availability of an organization's data. Data governance helps ensure data integrity and data security by defining and implementing policies, standards and procedures for data collection, ownership, storage, processing and use.

KEY COMPONENTS

- (i) Data Policies and Standards
- (ii) Data Ownership and Accountability
- (iii) Data Access Control
- (iv) Data quality Management
- (v) Data Security and Privacy

Q. What is Bias in AI?

AI bias, also called machine learning bias or algorithm bias, refers to the occurrence of biased results due to human biases that skew the original training data or AI Algorithm - leading to distorted outputs and potentially harmful outcomes.

When AI makes a mistake due to bias such as groups of people denied opportunities, misidentified in photos or punished unfairly - the offending organization suffers damage to its brand and reputation.

What Does it Mean for an AI model to be Biased?

- It learns from unbalanced or incomplete data.
- It reflects human prejudices present in the training data.
- It makes systematic errors against certain groups of people.

33. Comparison of Data Roles.

Role	Main focus	Key Skills	Common Tools
Data Analysts	Analyzing data to find insights	Data cleaning, visualization, Excel, Excel.	Microsoft Excel, Power BI, SQL
Data Scientist	-Building models & Predictions	Statistics, machine learning, Python	Python, Jupyter Notebook, Scikit-learn
Data Engineer	Building data pipelines & systems	Data architecture, ETL databases.	Apache Spark, Hadoop, SQL
AI Engineer	Developing AI/ML systems	Machine learning, deep learning, coding.	TensorFlow, PyTorch, Python.

34. Business Intelligence (BI)

BI refers to the technologies, tools and processes used to collect, analyze and present business data so organizations can make informed decisions.

BI turns raw data into meaningful insights through reports, dashboards and visualizations.

Popular BI tools and How They Help

① Power BI

- Microsoft tool for creating interactive dashboards and reports.
- Connects to multiple data sources

(ii) Tableau

- A powerful tool for data visualization and story telling
- Allows users to create interactive charts and dashboards

Why BI is Important in Decision Making.

- Faster Decisions
- Better Insights
- Improved Performance
- Data-Driven Culture

SECTION E: PERSONAL REFLECTION

35. The concept I found most interesting was data mining.
Data mining is the process of analyzing large datasets to discover patterns, trends and useful information.

What I found surprising is how businesses can use data mining to predict future behaviour, such as customer buying habits or trends. This shows how powerful data can be when it is properly analyzed.

36. One concept I found most difficult to understand was data modelling.
I used some websites and ChatGPT to understand it better.

37. I see data and AI playing a major role in both my future career and everyday life. I believe data will help me make better and faster decisions.